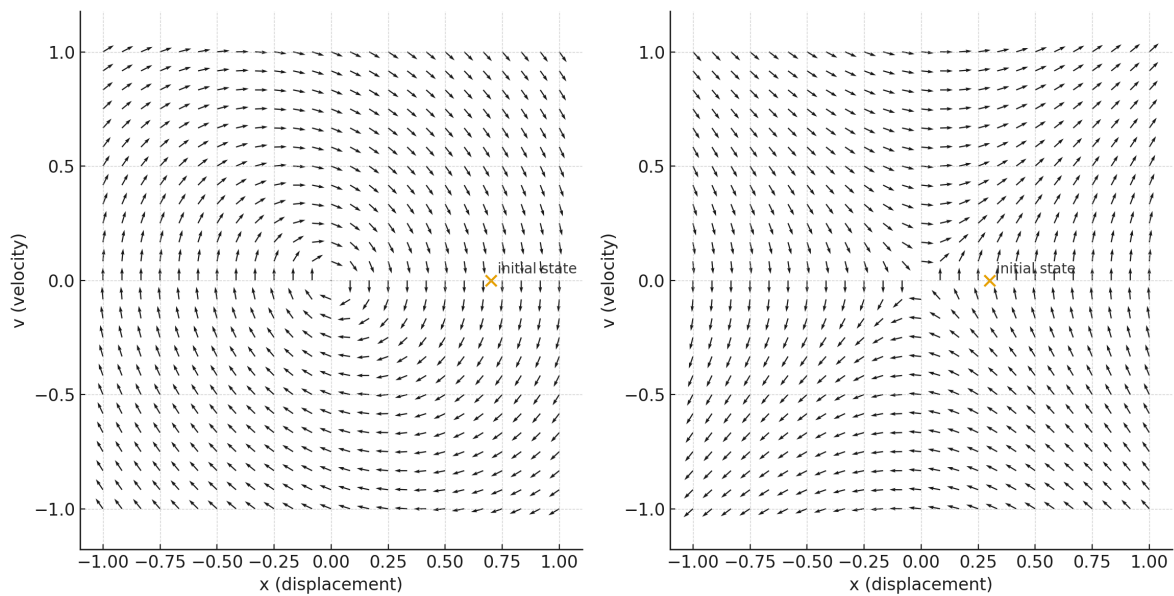


02YMECH - Homework 6

Assigned for the week of Oct 27, 2025

Questions

1 - Consider the following two diagrams:



Comment on the time evolution of the system in both the cases concisely. Which one would you classify as stable and which one as unstable? Justify your answer.

2- A point mass moves in the plane with polar coordinates (r, θ) relative to a fixed origin. Let

$$r(t) = r_0 + v_r t, \quad \theta(t) = \Omega t + \frac{1}{2}\alpha t^2,$$

where $r_0 > 0$, v_r , Ω , and α are constants.

- Write the general expressions for the velocity and acceleration in polar coordinates.
- Find the radial and angular components of the velocity and acceleration as functions of time.
- Compute the speed $v(t) = \|\mathbf{v}\|$ and its time derivative $\dot{v}(t)$. State the condition under which $\dot{v}$ is positive, negative, or zero.
- Find the time(s) t^* (if any) at which the radial component of acceleration a_r vanishes. State the condition on parameters for such a time to exist (with $t^* \geq 0$).